

Concepts of Routers and Switches

Identify the mode:

MAIN> This is a router or switch prompt that indicates: 1. The name of the router or switch is MAIN. 2. The > indicates that we are in USER mode. This is a look but don't touch or change anything. From our point of view this mode is not worth a whole lot.

MAIN>enable The enable command is used to change the router or switch from USER mode into PRIVILEGED mode.

MAIN# The # indicates that we are now in privileged mode. We can definitely look, but now we can change or modify the configuration.

A device that is first booted with no configuration will go from USER mode to PRIVILEGED mode simply by typing enable. A configured device will more than likely require a password to change between the two modes.

MAIN#configure terminal The command configure terminal or config t will allow a user in PRIVILEGED mode to transition in to a global configuration mode.

MAIN(config)# This GLOBAL mode allows you to make changes that will affect the behavior of the entire device.

You can go backwards through the modes by typing and entering EXIT or using the key combination Ctrl-Z.

Examples Below

```
Cisco 1841 (revision 5.0) with 114688K/16384K bytes of memory.
Processor board ID FTX0947Z18E
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1,
RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team
```

--- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no

Press RETURN to get started!

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname TrafficCop
TrafficCop(config)#enable password please
TrafficCop(config)#service password-encryption
TrafficCop(config)#enable secret please
TrafficCop(config)#line vty 0 5
TrafficCop(config-line)#password letmein
TrafficCop(config-line)#login
TrafficCop(config-line)#^Z
TrafficCop#
%SYS-5-CONFIG_I: Configured from console by console
show run
Building configuration...
```

```
Current configuration : 633 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname TrafficCop
!
!
!
enable secret 5 $1$mERr$GTblq7aeThiBpUjL7dCEp.
enable password 7 0831404B080A00
!
!
!
!
!
!
!
--More--
```

```
63488K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address: 0001.C743.0A9C
Motherboard assembly number: 73-5781-09
Power supply part number: 34-0965-01
Motherboard serial number: FOC061004SZ
Power supply serial number: DAB0609127D
Model revision number: C0
Motherboard revision number: A0
Model number: WS-C2950-24
System serial number: FHK0610Z0WC

Cisco Internetwork Operating System Software
IOS (tm) C2950 Software (C2950-I6Q4L2-M), Version 12.1(22)EA4, RELEASE SOFTWARE(fcl)
Copyright (c) 1986-2005 by cisco Systems, Inc.
Compiled Wed 18-May-05 22:31 by jharirba

Press RETURN to get started!

Switch>enable
Switch#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname MailMan
MailMan(config)#enable password please
MailMan(config)#service password-encryption
MailMan(config)#enable secret password please
MailMan(config)#line vty 0 15
MailMan(config-line)#pass letmein
MailMan(config-line)#login
MailMan(config-line)#^Z
MailMan#
%SYS-5-CONFIG_I: Configured from console by console
show run
Building configuration...

Current configuration : 1109 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname MailMan
!
enable secret 5 $1$mERr$JWuy1MseGVNApe.P7Rj2v.
enable password 7 0831404B080A00
!
!
spanning-tree mode pvst
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
--More--
```

